



MARIST
UNIVERSITY®

BIOL 392L 111
SP TOP:BIOL III
Fall 2026

Welcome to Microbial Bioinformatics! This course will introduce you to the foundational (and ever evolving) computational and bioinformatic resources that are necessary to analyze sequencing data. Given the rapid advancement of next-generation sequencing technologies, the way we are able to now easily access and conduct analyses has transformed the way we can explore genomic research questions in the realm of biology. We will be focusing on microorganisms such as bacteria and archaea and their proliferation to all types of "biomes" including the natural environment, human-systems, and built spaces or structures. A large focus of this course is how you can drive your own research questions during each step of our workflow to analyze whole microbial communities down to a single genome (even single gene or nucleotide if you wish). You will also learn how to assess genomic or other bioinformatic research from recent relevant literature. You will uncover the wonders of the genetic code that is packed inside sequencing data, and discover the exciting, yet complex, microbial world that surrounds you!

Instructor Information

Patrick Thieringer

Email: Patrick.Thieringer@marist.edu

Office Hours

T | 10:00-11:30am

W | 2:00-3:30pm

Optional Information

Schedule by appointment: <https://calendly.com/patrick-thieringer-marist/>

Course Information

BIOL 392L SP TOP:BIOL III

Credit Hours: 3

Meeting Location: DN238

DN238

Meeting Times: W 0930AM-1045AM
M 1100AM-1215PM

Course Description

“Special topics” courses serve as a vehicle by which a school may offer a topical or thematic study not included in the regular course offerings. The specific content is indicated when the course is listed in the schedule of classes.

Pre-Requisite/Co-Requisite

BIOL130 & 131 - General Biology I & II

BIOL 320 - Genetics

Course Learning Outcomes

COURSE GOAL I - Describe and perform all of the modern techniques, tools, and methods for analyzing next-generation sequencing data

SLOI-01 : Identify how to edit scripts and code with the command line to interact with data

SLOI-02 : Describe what each step of a modern sequence analysis pipeline is doing

SLOI-03 : Interpret the output of different steps of a bioinformatic workflows and the limitations associated with those steps

SLOI-04 : Critically analyze sequencing data and the surrounding implications or limitations of what can be inferred

COURSE GOAL II - Draw and diagram conceptual models from broad environmental system nutrient cycling to molecular level mechanisms

SLOII-01 : Develop an intuitive sense of how broad interactions between different

microbial members and environmental drivers are related

SLOII-02 : Explain the limitations or assumptions of a conceptual model to aid in the understanding of a system

SLOII-03 : Generate hypotheses based on unique or unexplained portions of a diagram

COURSE GOAL III - Gain the confidence to design thoughtful and crafted research questions

SLOIII-01 : Learn and practice how to ask questions that demonstrate active listening/reading of material or what could support a “next-steps” of a study

SLOIII-02 : Reason logically and develop the skills to think with quantitative and qualitative about datasets

SLOIII-03 : Develop a research voice through consecutive efforts to question individual datasets and practice sharing the results in front of their peers

COURSE GOAL IV - Gain literacy in coding and bioinformatics terminology to dissect relevant literature

SLOIV-01 : Develop a knowledge-base and vocabulary for ‘omics related research

SLOIV-02 : Distinguish the limitations of certain pipelines or workflows, and understanding what kinds of interpretations can be made

SLOIV-03 : Appreciate and promote the need for accessibility and reproducibility in bioinformatics

COURSE GOAL V - Demonstrate proficiency in communicating their science

SLOV-01 : Develop the skill to ask intriguing and well-crafted questions based on primary literature

SLOV-02 : Present findings in a clear manner that distinguishes between observations of results and interpretations of conclusions backed by similar research

SLOV-03 : Write coherent, technical papers in the same fashion of a scientific publication

Required Materials



Genomic Approaches in Earth and Environmental Sciences

ISBN: 9781118708248

Authors: Gregory Dick

Publisher: John Wiley & Sons

Edition: 1

Additional Information

This book is **NOT** necessary for this course. It is a nice companion for everything we will talk about, but not a requirement for the material and information we will cover.

Optional Resources

I will use Brightspace as the course website to post announcements, quizzes, course material, and supplemental information. All announcements will be sent to your email, so be sure to check your email at least once per day.

I will also create a Microsoft Teams group chat that the class will use as a way to communicate about anything coding related with their group or to the class. This is meant to mimic other forums that are hosted to debug or troubleshoot any issues while using a program or compare scripts. It is encouraged that you all work together as a class to fix any problems that may arise, as you will learn more about what a program is actually doing through troubleshooting than having the

instructor answer every time your script fails.

Teaching Methods and Learning Experiences

The format of this class is intended to be as applied as possible. In order to achieve this, we will be running the classroom as a lecture/lab hybrid format. There will be many class sessions where lecture will only span half of the allotted time so that the remainder of the period can be spent running code or analyzing other aspects of sequencing data.

To make the most of this format, many aspects of the classic lecture format will be "flipped" by providing reading material that you will spend time with outside of our slotted class period. This is where the iRAT/tRAT quizzes will play a role in determining how prepared we are as a class for the material we need to dive into.

You will be expected to bring your personal computer to class (if you do not have a personal computer, please come talk to me as soon as possible so that we can find one for you), and this does not include a tablet or other smart device. This is because you will actively be editing and running bioinformatics pipelines that we need to analyze the samples in this class. Your computer will be used to log onto a high performance computer (HPC) that has much greater memory and storage capacity than the average laptop can handle.

One final piece of advice about what to expect while working with the sequencing data in this class. The samples being provided to you come from real research experiments conducted around the world and use sequencing and metagenomic tools to answer only a facet of the questions that are available. As you design your own research project, it is likely you will stumble into aspects of the data that is weird or hard to explain - possibly even exciting more questions! To that end, a common theme or motto of this course is going to be: **ROLL WITH IT!** I will not know every aspect of the dataset that you are working with. You will be analyzing a set of samples in real time and we will be exploring it all together! So keep an open mind and a positive curiosity about the potential to explore a cool project from these data.

Course Grade Calculation

Assignment	Point Value
Participation & Attendance	100
iRAT/tRAT Quizzes (11 total)	220
Homework Assignments (2 Total)	40
Mini-Reports (8 Total)	240
Mini-Project Report	100
Mini-Project Presentation	100
Rough Draft & Conceptual Design of Final Project	50
Final Project Report	200
Final Project Presentation	150
Total	1200

Final Course Grade Scale

Grading Scale

Letter Grade	Percentage
A	100 - 95%
A-	94 - 90%
B+	89 - 87%
B	86 - 83%
B-	82 - 80%
C+	79 - 77%
C	76 - 73%
C-	72 - 70%
D+	69 - 65%
D	64 - 60%
F	59% or below

Graded Assignments

Participation & Attendance

Many of the learning objectives and computational lab components of this class revolve active learning components through live class demonstrations and discussion to apply bioinformatics research relevant to your work. Therefore, the participation component of your grade will require not only your consistent

attendance but also engaging in class discussions and course material. If you have an unexcused absence, you will forfeit your opportunity for participation points from that day. Participation points will not be adversely impacted by excused absences.

iRAT / tRAT Quizzes

These quizzes are a part of a teaching effort called Team Based Learning (TBL). You will learn more about this in greater depth at the beginning of the semester. Simply put, material will be provided as readings outside of the classroom (acting similar to a "flipped" classroom) which you will then be evaluated on at the beginning of the following class session. You will take the quiz first individually (individual Readiness Assessment Test) and then immediately again as a group (team Readiness Assessment Test).

This style of assessment is meant to test a student's readiness for the module we will be focusing on during any given week. This provides accountability on the individual to come prepared to help their group, but also so that students can come to class ready to apply their knowledge to interesting and applied problems. In a class without a true "lab" component, this is vital to making sure we can spend the appropriate class time on the different steps to process our sequencing data rather than be bogged down by lectures.

The individual and team based assessment will account for 60% and 40% of your final score for each quiz, respectively.

Homework Assignments

These are small assignments meant to prepare you for the content we are intending to cover in a following class. These assignments are graded for completion unless otherwise stated. This is to keep everyone on track with the syllabus schedule and make sure that the entire class is prepared to move forward with the next assignment.

Mini-Reports

Mini-Reports are short, open-ended assignments meant to apply your knowledge on a topic we recently covered in class. You will typically be asked to develop a research-style question regarding the tool or concept we have discussed for the week that you can answer with your current sequencing data results. A major goal of this class is to move beyond taking our results at face value, and to really understand what the data is informing us of. Many of these reports will require you to write a 500-750 word response with an accompanying figure/plot that you generate. By completing these consistently throughout the semester, these reports are also intended to help serve as a tool to aid your writing and synthesis skills - and to aid in your final project design.

Mini-Project Report

Different than the Mini-Reports, this Mini-Project report will be a more formal write-up (~1-2 pages) of all of the bioinformatic tools you have used halfway through the semester to aid you in your analyses. At the start of the semester, each group and individual will be assigned a "mystery" sample that you need to try and deconstruct to evaluate and hypothesize what kind of environment the sample was collected from. You will be expected to write this up as if submitting to a scientific journal and include relevant figures and citations. Further details of what needs to be included will be posted on Brightspace and discussed in class.

Mini-Project Presentation

You and your partner will be expected to give a presentation to the class regarding your "mystery" samples and what kind of environment you suspect you have been investigating. You will give a short presentation limited to a total of 10 slides that includes relevant data, figures, and information to support your claims. This is also an opportunity to demonstrate your proficiency in communicating and discussing all that we have been learning in class. A grading rubric and detailed information will be available on Brightspace.

Rough Draft & Conceptual Design of Final Project

Upon the completion of the mini-project presentations, we will reveal the

environment and relevant literature the data was acquired from for our "mystery" samples. From this halfway point in the the class, you will be expected to develop a research project/question that you would like to complete for the remainder of the semester. This should be a question not already discussed or analyzed within your assigned paper.

This assignment will serve as an opportunity to get structured feedback about your intended research project. This is meant to mimic what it's like to work with a research committee or review panel to help you reach a final project that is not only exciting but accessible with the resources available in this class. Examples of final project questions and the scope of the research will be discussed in detail during the class period.

Final Project Report

This report will coalesce your final research project in the style of a scientific paper. You should be able to clearly convey the hypothesis you are testing and the results of your findings. This report will be a culmination of your ability to dissect the relevant literature associated with your samples and environment, run the appropriate bioinformatic tools, and synthesize your findings. This report should be approximately 3-5 pages with a minimum of 4 figures. Detailed information will be discussed during class and also made available on Brightspace.

Final Project Presentation

You will give a final presentation regarding your final research project. This will take place instead of a final exam for this class. You will be expected to give a thorough 10-15 minute presentation that covers the question/hypothesis you intend to answer and explains your entire process of how you reached your conclusions. This should be a reflection of everything you have reported in your final project report, but should focus on providing clear visuals rather than text. You will also be expected to answer audience questions after your presentation that demonstrates your understanding of the system, methods, and research you have conducted. All the information necessary to prepare your presentation will be available on Brightspace and discussed during the final weeks of class to best prepare you for your talk.

Attendance Policy

If you're sick, please don't come to class and share your illness with everyone else. I will be as flexible as possible around in-class attendance, but communication is key if you will miss/think you may miss a class. Attendance is expected if you are healthy, but please respect your peers and everyone in the campus community by staying home (or wearing a mask in class) if you're sick.

Lecture: You are encouraged to attend all class meetings. All content is available on the course website and I'm happy to address questions during office hours if you cannot make it to class. Additionally, most assessments can be delivered via Brightspace, so you will be able to turn in work, take quizzes, etc. from outside of the classroom and not fall behind if you are out of class for a few days.

Academic Support

Marist University provides a range of resources to help you get the most from your educational experience. These resources include but are not limited to tutoring, writing assistance, advising, proofreading assistance, and counseling. A full list of these resources can be found at this link. <https://www.marist.edu/academic-resources>

Technical Support and Support Links

Classrooms and Labs support services are managed through the IT Help Desk and Media Center. Computer related problems should be reported by calling the Marist University IT Help Desk at (845) 575-4357 (xHELP) or email: helpdesk@marist.edu. Problems with overhead projectors or other media related equipment should be reported to the Media Center Help Desk at (845) 575-3635 (x3635) or email: media@marist.edu.

To access Brightspace, visit: <https://brightspace.marist.edu/d2l/home> and click on "Marist SSO" to login directly with your Marist credentials. For helpful Brightspace tool guides and tutorials, visit: <https://my.de.marist.edu/teaching-technologies>

Statement on Academic Honesty

Marist University is a learning community dedicated to helping students develop the intellect, characters, and skills required for enlightened, ethical, and productive lives in the global community of the 21st century. Students are expected to pursue excellence in their education while being honest about their work and fair to other members of the learning community. All work presented to instructors for evaluation must reflect their own ideas and effort and must properly acknowledge any contributions of others. Students should expect this honesty and fairness in others as well. As members of the Marist learning community, all students should adhere to the principles of academic integrity as set forth in the Marist Academic Integrity Policy.

<https://www.marist.edu/student-life/community/community-standards/academic-integrity-policy>

AI Statement

Structured Use for Academic Integrity: Generative AI tools (e.g., text and image generators like ChatGPT and DALL-E) may not be used to complete assignments unless explicitly permitted. Unauthorized use of AI tools will result in a violation of the academic integrity policy. If AI use is allowed for certain assignments, students must document its role in their work.

Detection Software Statement

Use of software that detects plagiarism, including AI generated content, is mandated in this course. Copyleaks is a service used by Marist University faculty to compare a student's work with its very large database of sources, student papers from other institutions, and the like, to check for originality. Work submitted to Copyleaks will be used only for the purposes of assessing originality, and will not be shared beyond Copyleaks or used for any other purpose. Students must submit all assignments to Copyleaks through regular Brightspace submission process. If your instructor enables Copyleaks, there will be an End

User License Agreement (EULA) that you must accept. It will appear to you on the assignment page before the area where you submit the assignment. You can view and accept the EULA by clicking on the link and then clicking "Finish" to accept the terms of use. Students who wish to remove their personal identifying information (name, student identification number, etc.) from the submitted file may do so but must notify their professor ahead of the submission. Work submitted through Brightspace in this course will not be reviewed by the Professor or maintained by the University unless and until the Copyleaks process is completed.

For more information about Copyleaks visit: <https://my.de.marist.edu/copyleaks>

Accommodations and Accessibility

From the Office of Accommodations and Accessibility

Students with disabilities who believe they may need accommodations in this class are encouraged to contact the Office of Accommodations and Accessibility at (845) 575-3274, Donnelly Hall 226 or via email at accommodations@marist.edu as soon as possible to better ensure that such accommodations are implemented in a timely manner. Marist's guidelines for instructors to comply with the Americans with Disability Act (ADA) are located here: <https://www.marist.edu/student-life/community/accommodations-accessibility/guidelines-instructors>

Your Privacy Rights - Family Education Rights and Privacy Act

The Family Educational Rights and Privacy Act (FERPA) (20 U.S.C. § 1232g; 34 CFR Part 99) is a Federal law that protects the privacy of student education records. Marist University collects and retains data and information about students for designated periods of time for the expressed purpose of facilitating the student's educational development. The University recognizes the privacy rights of individuals in exerting control over what information about themselves

may be disclosed and, at the same time, attempts to balance that right with the institution's need for information relevant to the fulfillment of its educational missions. As employees of the University, faculty, administration and staff all have some access to student record information. Please, visit <https://www.marist.edu/academics/registrar/ferpa> for more information on how Marist University protects your privacy rights.

Copyright Information

Materials in this course may be subject to copyright protection.

Title IX Statement

Marist University is committed to providing a safe learning environment for all students. If you or someone you know has experienced sexual harassment, including sexual assault, dating or domestic violence, or stalking, support is available. Please contact the Title IX Office at titleix@marist.edu or (845) 575 - 3799 or visit www.marist.edu/title-ix to file a report. Please be aware that faculty and staff are required to disclose incidents of sexual harassment or other potential violations of the Marist University Discrimination, Harassment, and Sexual Misconduct Policy to the Title IX Office. To speak to a confidential resource who does not have this reporting responsibility, contact Counseling Services at (845) 575 - 3314, Health Services at (845) 575 - 3270, or Department of Spiritual Life & Service at (845) 575 - 3000 (x2275). For more information about reporting options and resources at Marist University, please visit <https://www.marist.edu/title-ix>

Community and Belonging Statement

The university's academic mission is immeasurably enriched by students with diverse experiences. Our finest efforts as intellectual beings heavily rely on the exchange of ideas. Interactions in our classrooms among persons and groups with diverse backgrounds, ideologies, and experiences facilitate these efforts by allowing us all to be more reflective about the varied historical and social contexts in which we work and learn. For faculty and students to continue being leaders inside and beyond academia, we must ensure that we consider the diversity of all

who comprise our communities and foster a climate in which those diverse influences are respected and valued. In this course, we will challenge each other's thinking while working collaboratively to ensure that the classroom is a space of safety and bravery. Our classroom offers an environment where individuals of varying opinions, experiences, and backgrounds are able to be free to learn without fear of being silenced. Evidence of these efforts will manifest in readings, lectures/class discussion, seminars, and group projects. Aspects of diversity include, but are not limited to, race, ethnicity, color, nationality, sex, gender, gender identity, gender expression, class, sexual orientation, religion, age, ability, and veteran status. Students who would like to be identified in a manner other than what is indicated on the course roster can contact me privately via phone, email, web conference or face-to-face meeting to indicate name, pronoun and any other preferences they may have.

Additional Course Policies

LATE POLICY

Most assignments (not exams) have a penalty-free 48-hour grace period from the submission deadline. Any assignments which do not have this grace period will be clearly indicated on the class website.

- Questions about assignments will not be answered during the grace period, or after 5pm on the original due date of the assignment.
- Assignments will NOT be accepted 7 days after the due date and you will receive a zero for that assignment

Following the 48-hour grace period:

- Assignments submitted up to 24 hours after the end of the grace period will incur a 10% penalty.
- An additional 10% penalty will be applied for each additional 24hr period.

EXCUSED ABSENCE POLICY

The following constitute the only valid excuses for missing class, quizzes, exams, assignments or in-class activities:

1. Death in the immediate family. Present a note from the Center for Advising & Academic Services.
2. Officially-approved College activities such as intercollegiate athletic events or theatrical productions. Present a note from the appropriate College official stating that your participation in this activity prevented you from being in class at the exact date and time it was held.
3. Illness. If you're sick or think you might be getting sick, STAY HOME. As noted above, all assessments and assignments can be completed online, so you are expected to stay on track with quiz dates, assignment due dates and other components throughout the semester whether you are in the classroom or remote.
4. Debilitating illness. If you are too ill to complete quizzes/assignments/etc. on time, I will require documentation stating that the illness prevented you from participating in class remotely at the exact date and time that the quiz/assignment and/or laboratory was held. If you seek medical attention at Marist Health Services, you must provide health service with documented authorization allowing me to verify your illness or injury. A note that simply states that you were under the care of a physician, or had an appointment with a doctor, is not sufficient.

If an absence is excusable and of sudden or unpredictable nature, I still require notification as soon as possible (at the latest: on the day you return to class; preferably prior to that date, via email) in order to reschedule missed assessments. Please note that it is your responsibility to schedule a make-up when you return to class. If a make-up is not rescheduled within one week of your return to class, you will have lost your opportunity to do so.

Course Calendar

Week	Date	Topic	HW & Assignments	Quiz Module
1	31-Aug	Exploring the Microbial World & Syllabus Breakdown	HW #1 - Microbes in your favorite environment	
	2-Sep	Sequencing Revolution & Application of 'Omics to Environmental/Built Systems		
2	7-Sep	Labor Day		
	9-Sep	Sequencing Overview & FASTA/FASTQ Files	Get Cyberduck installed & have Terminal running	What is iRat/tRat?
3	14-Sep	How to use an HPC & Practice with the CLI	HW #2 - Directory Management	
	16-Sep	Introduction to BLAST and other Databases & Keeping a Coding Notebook		HPC & BLAST/Databases
4	21-Sep	Further Intro to HPC - Start Analysis of Mystery Samples		
	23-Sep	Quality Control of Raw Sequencing Files	Mini-Report #1 - FastQC Results	QC of Sequencing Results
5	28-Sep	Assembly Overview		Assembly & Assessing Stats
	30-Sep	Assessing Assembly Statistics & Quast	Mini-Report #2 - Quast Results	
6	5-Oct	Mapping	Mini-Report #3 - Read Mapping Experiment	Mapping & Binning
	7-Oct	Binning		
7	12-Oct	Genome Quality Evaluation		Genome QC & Taxonomy
	14-Oct	Taxonomy & Diversity	Mini-Report #4 - Genome QC and Diversity Statistics	
8	19-Oct	Office Hours & Prep for Mini-Project Presentations		
	21-Oct	Mini-Project Presentations	Submit Mini-Project Reports	
9	26-Oct	Phylogenomics I - SCG Trees	Select MAG	Phylogenomics
	28-Oct	Phylogenomics II - Single-Gene Trees	Mini-Report #5 - Phylogeny of Selected MAG	
10	2-Nov	Functional Annotation		Functional Annotation & Metabolic Pathways
	4-Nov	Metabolic Pathways & Contextualizing Biogeochemical Cycling	Mini-Report #6 - Metabolic Potential of Environment	
11	9-Nov	Pangenomes & Anvi'o		Pangenomes & Data Visualization
	11-Nov	Data Visualization & Exploration	Mini-Report #7 - Evaluate Figure from Paper	
12	16-Nov	Antimicrobial Resistance Genes & Secondary Metabolites		ARGs & Secondary Metabolites
	18-Nov	Phages, Plasmids, & Mobile Genetic Elements	Draft of Final Project	
13	23-Nov	Case Studies & Reproducibility of Metagenomics Data	Mini-Report #8 - Reproducibility & Accessibility of Final Project	Reproducibility of Metagenomic Data
	25-Nov	Thanksgiving		
14	30-Nov	Open Lab (Other topics discussed upon interest)		
	2-Dec	Open Lab		
15	7-Dec	Open Lab		
	9-Dec	Open Lab		
	14-Dec	8:00 am - Final Presentations	Final Project Report	